

8.1 How can cost and availability of specific materials and/or system components affect selection when designing?

a.	The significance of the cost of specific materials and/or system components in relation to commercial viability, different stakeholder needs and marketability.	
b.	How to calculate the quantities, cost and sizes of materials and/or system components required in a design or product, being able to apply this to different applications.	

Mathematical skills in the Principles exam

It is a requirement of this qualification that a minimum of 15% of the written exam assesses the use of mathematical skills at a level of demand which is not lower than that expected at Key Stage 3.

The required mathematical skills to be tested in the examination are listed in the tables below. In order to support the application of these skills, examples of how the skills may be applied to design and technology and may be assessed in an exam are provided. The examples given are not exhaustive. The content in the specification with the Maths symbols also indicates and supports possible content for application.

Where mathematical skills are learnt through 'in-depth' areas, learners should be able to apply this learning to other broader contexts.

Learners are permitted to use a scientific or graphical calculator for this written examination. Calculators are subject to the rules in the document *Instructions for Conducting Examinations* published annually by JCQ (www.jcq.org.uk). However, it is expected that how a calculation was reached should be shown by a learner in their responses.

M1 Arithmetic and numerical computation

Required skills

Examples of application to Design and Technology

a. Recognise and use expressions in decimal and standard form.	- Understand the standard application of metric units used in Design and Technology and apply these appropriately using standard form (also be aware that some measurements commonly retain the use of imperial units). - Use decimal and standard form appropriately when using units of mass, length, time, money and other measures. - Use and apply standard form when calculating quantities of materials, cost and sizes.
b. Use ratios, fractions and percentages.	- Understand and use ratios in the scaling of drawings. - Understand and apply fractions and percentages when analysing data, survey responses and user questionnaires given in tables and charts. - Calculate percentages e.g. with profit, waste saving calculations or comparing measurements.
c. Calculate surface area and volume.	- Calculate the surface area to determine quantities of materials (where dimensions are given). - Calculate the volume of cuboids, simple and composite shapes (where dimensions are given). - Apply tolerances to material dimensions and quantities.

M2 Handling data

a. Presentation of data, diagrams, bar charts and histograms.	- Construct and interpret frequency tables, pie charts and bar charts from given data sources. - Present given information to accurately represent performance over time.
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<i>Required skills</i>		<i>Examples of application to Design and Technology</i>
a.	Plot draw and interpret appropriate graphs.	- Plot or draw graphs from given information such as performance data and client survey responses. - Analysis graphs to interpret and extract information.
b.	Translate information between graphical and numeric form.	Extract information from technical specifications and graphical sources to understand instructions or requirements.
M4 Geometry and trigonometry		
a.	Use angular measures in degrees.	- Know the basic properties of isosceles, equilateral and right angle triangles. - Understand the basic rules of angular calculations and trigonometry to support accurate marking out. - Demonstrate accuracy in measurement and marking out of angles. - Understand symmetry to create tessellated patterns to minimise waste.
b.	Visualise and represent 2D and 3D forms including two dimensional representations of 3D objects.	- Graphical presentation of designs to communicate intentions to others. - Transfer information from 3D objects with given dimensions into accurate 2D representations. - Interpret information to accurately present isometric drawings.
c.	Calculate areas of triangles and rectangles.	- Calculate areas to determine quantities of materials (where not all dimensions are given). - Calculate area to determine area scale factor application.
d.	Calculate the surface area and volumes of cubes.	- Calculate the overall surface area of cuboids to determine quantities of material. - Calculate the volume of cuboids to determine suitability of objects fitting into a space (where not all dimensions are given). - Calculate the volume of cuboids to determine volume scale factor application.

Further details on the forms of assessment for examinations can be found in Section 3a.

The mathematical requirements are mapped out against the OCR GCSE (9–1) in Mathematics in Section 5c.